# LITERATURE-IMPLIED ANSWER TO QUESTION QUEST:KPT IN ARXIV:2408.12755

FA BANACH RUN PACKET

## Status

This packet records a literature-implied full answer to Question `quest:kpt` from Cuth–de Rancourt–Doucha, *Guarded Fraisse Banach spaces*, arXiv:2408.12755. The source asks:

> Fix $1 \le p, q < \infty$. Is the group $\mathrm{Iso}(L_p(L_q))$, endowed with the strong operator topology, extremely amenable? If not, does it have a metrizable universal minimal flow?

The answer is affirmative for every $p, q$. Consequently the universal minimal flow is a singleton.

The relation is not stated in the source and was not found as an explicit later answer in the cheap index or exact web searches. It follows by combining standard scalar $L_p$-isometry extreme amenability, Pestov–Schneider's measurable-map theorem, and the Guerre–Raynaud structure theorem for isometries of $L_p(L_q)$.

## Theorem

Let $1 \le p, q < \infty$, and let $L_p(L_q)$ mean

$$L_p\big([0,1]; L_q([0,1])\big).$$

Then $\mathrm{Iso}(L_p(L_q))$, with the strong operator topology, is extremely amenable. Equivalently, every continuous action of this group on a compact Hausdorff space has a fixed point.

## Proof

We use three external facts.

First, the scalar isometry group $\mathrm{Iso}(L_r)$, with the strong operator topology, is extremely amenable for every $1 \le r < \infty$: for $r = 2$ this is the Gromov–Milman theorem for the infinite-dimensional orthogonal/unitary group, and for $r \ne 2$ this is the Giordano–Pestov theorem, as quoted in the source discussion of Question `quest:kpt`.

Second, Pestov–Schneider prove that for every topological group $G$,

$$\begin{aligned} G \text{ amenable} &\iff L^0(G) \text{ amenable} \\ &\iff L^0(G) \text{ extremely amenable} \\ &\iff L^0(G) \text{ whirly amenable}, \end{aligned}$$

where $L^0(G)$ is the group of strongly measurable maps $[0,1] \to G$, modulo equality a.e., with convergence in measure. Since extreme amenability implies amenability, this applies to $G = \mathrm{Iso}(L_q)$, giving that $L^0(\mathrm{Iso}(L_q))$ is extremely amenable.

Third, for $p \ne q$, the Guerre–Raynaud isometry theorem for vector-valued $L_p$-spaces identifies the strong-operator-topology group $\mathrm{Iso}(L_p(L_q))$ with the natural measurable-field extension

$$1 \longrightarrow L^0(\mathrm{Iso}(L_q)) \longrightarrow \mathrm{Iso}(L_p(L_q)) \longrightarrow \mathrm{Iso}(L_p) \longrightarrow 1.$$

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Concretely, every surjective linear isometry of $L_p([0,1];L_q)$ is a base $L_p$-isometry together with a strongly measurable field of fiber isometries of $L_q$. The source paper itself invokes Guerre–Raynaud, Theorem 5.1 and Proposition 1.6, in this form in the proof of Proposition `prop:LpLqNotAuH`, where an isometry $T \in \mathrm{Iso}(L_p(L_q))$ is represented by an isometry $S \in \mathrm{Iso}(L_p)$ and a measurable field $(T_t)_{t\in[0,1]}$, $T_t \in \mathrm{Iso}(L_q)$.

It remains only to use the elementary extension permanence of extreme amenability. If $N \triangleleft G$ is a closed normal subgroup, $N$ is extremely amenable, and $G/N$ is extremely amenable, then $G$ is extremely amenable: for any compact $G$-flow $K$, the fixed-point set $K^N$ is nonempty and compact; it is $G$-invariant, and the induced $G/N$-action on $K^N$ has a fixed point.

For $p \neq q$, apply this lemma to the exact sequence above. The kernel $L^0(\mathrm{Iso}(L_q))$ is extremely amenable by Pestov–Schneider, and the quotient $\mathrm{Iso}(L_p)$ is extremely amenable by Gromov–Milman/Giordano–Pestov. Therefore $\mathrm{Iso}(L_p(L_q))$ is extremely amenable.

For $p = q$, the Fubini isometry identifies

$$L_p([0,1];L_p([0,1])) \cong L_p([0,1]^2),$$

which is isometric to the usual atomless separable $L_p$-space. Hence $\mathrm{Iso}(L_p(L_p)) \cong \mathrm{Iso}(L_p)$ is extremely amenable by the same scalar theorem.

This covers all $1 \leq p, q < \infty$.

## Scope and Limitations

The answer is full for Question `quest:kpt`: the “if not” alternative does not arise, and the universal minimal flow is the one-point flow.

This packet does not address the other open questions in Section 7 of arXiv:2408.12755. It also does not claim novelty of the component theorems; the contribution is the direct identification that these known results answer the source question.

## Search and Provenance

Cheap-index searches for arXiv:2408.12755, *guarded Fraisse*, $\mathrm{Iso}(L_p(L_q))$, and *extremely amenable* found only the existing failed attempt packet for the $L_p(E)$ route and no promoted solution packet. Exact web searches for the source title, guarded Fraisse Banach spaces, $L_p(E)$ guarded Fraisse, and $\mathrm{Iso}(L_p(L_q))$ extremely amenable did not surface a later explicit answer. Thus this is recorded as an agent-identified literature implication rather than a later-paper “already answered” match.

## References

## References

[1] M. Cuth, N. de Rancourt, and M. Doucha, *Guarded Fraisse Banach spaces*, arXiv:2408.12755.

[2] S. Guerre and Y. Raynaud, *Sur les isometries de $L^p(X)$ et le theoreme ergodique vectoriel*, Canad. J. Math. 40 (1988), no. 2, 360–391.

[3] M. Gromov and V. D. Milman, *A topological application of the isoperimetric inequality*, Amer. J. Math. 105 (1983), no. 4, 843–854.

[4] T. Giordano and V. Pestov, *Some extremely amenable groups related to operator algebras and ergodic theory*, J. Inst. Math. Jussieu 6 (2007), no. 2, 279–315.

[5] V. G. Pestov and F. M. Schneider, *On amenability and groups of measurable maps*, arXiv:1701.00281.